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INFANT PERCEPTION AND COGNITION

An Evolutionary Perspective on Early Learning

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The young infant faces a seemingly daunting task in making sense of the world. The environment consists of a huge array of objects, entities, and events that possess and exhibit both considerable regularity and variability. For example, all things in the world tend to possess shape and size constancy, move as bounded wholes, follow the laws of physics, and possess clusters of correlated features. Yet some things move and act on their own volition, cause other things to move, and have goals and intentions, whereas others are caused to move and do not possess psychological states. Similarly, some things are potential threats, namely, certain animals, heights, and other humans, whereas others are benign or fundamental for the infant's survival.

The adaptive problems facing the human infant's perceptual and cognitive systems, and presumably that of the young of other species, are therefore to make sense of this multitude of information rapidly and in a veridical manner. Not only must the young of the species parse the array into individual objects, entities, and events, but they must sort these things into appropriate, sometimes context-dependent groups, recognize them when they are present, and learn which properties (e.g., agency, goal-directed, dangerous) are typical of which ontological kinds. Failure to accomplish this task efficiently, rapidly, and accurately would have dire consequences for the infant's probability of survival not only in the first years of life but beyond, into later childhood and adulthood.

Somewhat surprisingly, a scan of the developmental literature reveals that very few researchers consider an evolutionary approach in the study of early perceptual and cognitive development. A keyword search on PsychINFO—one of the most extensive databases available to psychologists—for infancy research on "perception" or "cognition" shows that over 3,000 articles or chapters have been published on each topic. However, the same database reveals that fewer than 100 articles or chapters have been published on infant perception and cognition as they relate to evolution, despite the fact that there are over 20,000 hits for the term "evolution" alone. What can explain this lack of focus on the evolutionary basis of perception and cognition?

There are a number of plausible answers to this question, but I propose just two here. First, many developmental psychologists, like psychologists in other fields, are unaware about evolutionary theory and have little exposure to it during their training. Second, developmental researchers are generally concerned with aspects of ontogenetic development rather than phylogenetic development; that is, the focus for those who study infant perception and cognition has usually been on what develops and when. For example, recent research on early categorization has shown that infants as young as 3 months of age can categorize cats as different from dogs and do so on the basis of surface properties such as facial features (Quinn, Eimas, & Rosenkrantz, 1993; Quinn & Eimas, 1996). However, few researchers discuss such important findings in light of the "how" and "why" questions: How do infants form category representations for diverse objects or, in other words, which mechanisms underlie such abilities? Why did these mechanisms, and not others, evolve? What adaptive problem is solved by the early onset of categorization? (Note that Quinn and colleagues do in fact consider such issues.) These kinds of questions are common to those who adopt an evolutionary approach to the study of the mind and behavior, but they are far from common in the developmental literature that addresses early perceptual and cognitive development. One of my goals in this chapter is to show that such an approach is necessary if researchers are to gain a complete understanding of development in these, and other, areas. Without it, researchers can appreciate only one part of the puzzle—what perceptual and cognitive abilities develop and when do they emerge—and they cannot hope to answer crucial questions relating to how these abilities develop and why they exist at all.

Although only a few researchers have explicitly adopted an evolutionary approach to perceptual and cognitive development, a number of theoretical views could be described as falling within the tenets of such an approach. These views, recently labeled as *core knowledge theories* (Siegler, DeLoache, & Eisenberg, 2003), have in common the view that the adaptive solution for rapid and efficient perceptual and cognitive development is to outfit infants with innate knowledge or specialized learning mechanisms in certain domains (e.g., Baillargeon, 1995; Carey & Spelke, 1994; Gelman & Williams, 1998; Mandler, 1992). (The term "innate" is used throughout the chapter to refer to mechanisms or content that "arise as a result of interactions that occur within

the organism itself during ontogeny" [Elman et al., 1996, p. 22]). The core knowledge position is summarized succinctly by Gelman and Williams (1998, p. 600):

> The brain is no less a product of natural selection than the rest of the body's structure and functions. . . . Hearts evolved to support the process of blood circulation, livers evolved to carry out the process of toxin extraction, and mental structures evolved to enable the learning of certain types of information necessary for adaptive behavior.

In line with evolutionary psychologists who focus on aspects of adult behavior (e.g., Cosmides & Tooby, 1994; Buss, 2004), supporters of this view therefore stress that dedicated psychological mechanisms have evolved to solve adaptive problems faced by infants throughout human history. These perceptual and cognitive mechanisms have evolved, it is argued, because aspects or domains of the environment have been stable over time, and it is ontologically advantageous for the developing mind to be "prepared" to process inputs from such domains. Examples of environmental stability include the properties of objects, such as object boundedness or size constancy; the laws of physics, such as gravity and support; and aspects of mathematics, such as addition and subtraction. Because input in these domains has remained consistent over time, for instance, the force of gravity has been in effect on earth as long as life has existed on it, it is adaptive for organisms to come into the world expecting such consistencies to exist. According to this view, then, the most evolutionarily viable solution for making sense of the vast array of information in the world is for the brain to possess different mechanisms or modules that process different kinds of information.

In contrast to this view, a number of researchers have suggested that cognitive, and to some extent, perceptual development is grounded in a number of domain-general mechanisms that process a broad array of information (Cohen & Cashon, 2003; Johnson, in press; Quinn & Eimas, 1996; Rakison, 2003; Roberts, 1998; Smith & Heise, 1992). The term "domain-general mechanisms" is used here to refer to learning processes that can operate on a wide variety of inputs—for example, auditory, visual, and tactile stimuli—and that are not context- or information-specific. These mechanisms include, but are not limited to, associative learning, classical and operant conditioning, habituation, and imitation. Thus, the claim is that infants learn about varied aspects of the environment such as gravity, addition, and the properties of objects through the same basic learning mechanisms that are sensitive to, and are able to encode, regularities in the environment. How can such relatively simple and unrefined processes underpin the acquisition of a wide variety of complex information? Turning the evolutionary approach on its head somewhat, this view implicitly assumes that the presence over phylogenetic time of regularities in the environment means mechanisms that are sensitive statistical learners can be used to learn about those regularities. In other words, there is

sufficient structure and regularity in the world across a wide range of domains that domain-general mechanisms can account for a good deal of early perceptual and cognitive development.

Although few theorists discuss this approach from an evolutionary viewpoint, I believe that it is important to note that such general learning mechanisms would have emerged to solve the adaptive problem of making perceptual and cognitive sense of the world; that is, domain-general processes, just like domain-specific ones, can be viewed as evolution's solution to learning. This argument goes against a number of intrinsic views within evolutionary psychology, most notably that an evolved psychological mechanism implies modularity and domain specificity. However, I suggest that it need not be the case that an evolved mechanism is a specialized one. The position I adopt in this chapter is that it is viable for humans, as well as animals, to possess both domain-general and domain-specific mechanisms for learning. Domain-general mechanisms, I suggest, operate for information that is highly structured in the environment and that has remained consistent over phylogenetic time. For example, such mechanisms are sufficient for infants to learn about the physical laws that govern objects in the world, develop object concepts for animals, and perceive speech components of one's native tongue. If domain-general mechanisms were the earliest to evolve and were sufficiently robust for rapid and veridical learning in these areas, there would be little selection advantage for the emergence of other specialized processes. Domain-specific mechanisms, in contrast, operate for information that is not as highly structured in the environment and/or that is relevant for survival and reproduction. For instance, heights, strangers, and snakes do not in and of themselves emit information that informs infants that they are to be avoided, yet failure to steer clear of them could result in injury or death. Some basic representational content could be prewired for such stimuli, what has been called minimal information-specific predispositions (Elman et al., 1996), as well as other areas of perception such as Gestalt principles and orientation to faces.

A similar view has recently been proposed by Geary and Huffman (2002), who provide an evolutionary framework for understanding brain and cognitive plasticity. In this context, they suggest that different kinds of plasticity have evolved as a result of different kinds of variability in the input. Thus, they argue that "information patterns that covary with survival and reproductive outcomes and that are invariant across generations favor the evolution of inherent, gene-driven constraints on brain organization and cognitive functions. Variable information patterns, in contrast, favor the evolution of brain and cognitive systems that are open to experience-driven modifications in these domains" (p. 677). Geary and Huffman call this approach *soft modularity* and suggest that domain-specific modules may have evolved to deal with the faces of conspecifics, the detection of favorably colored foods, and perhaps the acoustical patterns of conspecifics' vocalizations.

Although the framework presented in this chapter and that presented by Geary and Huffman (2002) have in common the notion that domain-general

and domain-specific mechanisms can operate in tandem, it is important to note one key difference. Geary and Huffman suggest that regularity in the environment led to the evolution of mechanisms or modules to process this regularity. For example, they argue that humans may possess a module for understanding the physical world. In contrast, the view I suggest here is that domain-specific modules were unnecessary for many aspects of the environment that remained unchanged over phylogenetic time and that were not crucial to early survival. Infants do not, for example, need to have a naive understanding of physics to survive the first months of life, and the structure in physical events would allow more general learning mechanisms to process this information. One could argue that selection cannot, by definition, operate on environments that have no consequences for fitness, so selection for domain-general mechanisms for information irrelevant to fitness cannot occur. However, I suggest, albeit speculatively, that domain-general mechanisms were among the first phylogentically to evolve, because they represented a flexible solution to a multitude of adaptive problems. Over time, more specialized mechanisms have evolved to solve domain-specific problems; yet domain-general mechanisms remain the primary foundation for learning.

With this framework in mind, in the following sections I examine the evidence from the perceptual and cognitive development literature that learning in different domains is underpinned by domain-specific or domain-general mechanisms. Practically speaking, it is extremely difficult to determine whether one kind of mechanism or another is at work. How could one show, for example, whether specialized mechanisms or associative learning account for how infants learn about the motion properties of objects and entities in the world? Or that the expectation that things fall until they make contact with a horizontal surface is innately specified rather than learned through domain-general processes? In a few rare cases, evidence with newborns can speak directly to this issue (e.g., Slater, 1989; Morton & Johnson, 1991) by showing the presence of a behavior at birth (the absence of a behavior at birth is not evidence that it is not innately specified). Also, converging data suggest that specific behaviors develop with universality and with very little direct experience (e.g., Ainsworth, Blehar, Waters, & Wall, 1978; Gibson & Walk, 1960). One could also argue that development that occurs in a precise, regular, and predictable fashion despite considerable individual differences in childhood experiences is indicative of evolved, domain-specific mechanisms. But in the majority of areas covered in this chapter, no evidence exists that falls under these headings. For example, there is only limited evidence with newborns, and individual differences in experience are rarely recorded. To address this problem, I apply a relatively simple rule wherever appropriate: If domain-general mechanisms are sufficient to explain infants' behavior, this will be taken as suggestive, at least, that such mechanisms are present. Not surprisingly, this approach is far from perfect. Applying parsimony as a criterion is a good rule of thumb, but it can lead to both Type 1 and Type 2 errors. Nevertheless, it is the approach adopted here, and I endeavor to apply it with care.

PERCEPTUAL DEVELOPMENT IN INFANCY

Given the space limitations on a chapter of this sort, I confine my discussion in this section to fundamental aspects of perceptual development, and in particular I focus on early visual development. This may seem like a strange choice, because vision is perhaps the least well developed of the senses at birth, due most likely to the lack of visual input in the womb. And clearly, perceptual development of the other sensory modalities is crucial to early survival, and research on taste, smell, touch, and hearing are rife with evidence of phylogentically evolved adaptations. Newborns prefer the taste and smell of sweet nourishment over savory nourishment (and can discriminate a number of tastes), most likely because of the need for the energy given by sugar. Newborns—specifically those that are breast-fed—identify and prefer the smell of their mother's scent over that of a stranger (Porter, Bologh, & Makin, 1988). Newborns possess touch-based reflexes that facilitate eating, avoidance of stimuli (e.g., coughing, sneezing, blinking), and even clinging to a caretaker. And newborns can recognize and prefer the sound of their mother's voice to that of a stranger, though quite possibly this is the result of more general learning processes in operation while *in utero* (DeCasper & Fifer, 1980).

If there is such clear evidence for evolved mechanisms for survival in these areas of perception, why, one might ask, would this section focus on visual perception? The most compelling, and obvious, reason is that the visual array is the primary channel of information input for humans. To survive and reproduce, we must, among other things, identify caregivers, find and attract potential mates, and avoid predators. We must find food and shelter, and we must use tools and objects appropriately. Although the other sense modalities are involved in all of these actions, I suggest that it is visual information that is crucial for their success; that is, human and nonhuman primates are fundamentally visual creatures: Members of the opposite sex, caretakers, predators, food, shelters, and tools are easily identified by their visual appearance. A second reason, and one that stems from the importance of visual perception throughout the lifespan, is that research on this area is more extensive and in-depth than that on the other sensory modalities. Consequently, in the following section I focus on depth perception, shape, size, and form perception, face perception, movement and object perception, and causal perception.

Depth Perception

A seminal study by Gibson and Walk (1960) with the visual cliff, which was one of the first specifically to examine perceptual development, is often cited as evidence for an evolved perceptual mechanism. In the study, 6- to 14-month-old infants rarely crawled across an apparent "deep" illusory drop when called by their mothers. At the very least, this suggests that by 6 months of age infants perceive depth. However, the authors suggested a more controversial conclusion:

> Common sense might suggest that the child learns to recognize falling-off places by experience—that is, by falling and hurting himself. But is experience really the teacher? Or is the ability to perceive and avoid a brink part of the child's original endowment? (Gibson & Walk, 1960, p. 64)

Evidence on depth perception with infants considerably younger than 6 months of age could be taken as evidence to support this position. Infants as young as 1 month of age blink at objects that appear to approach them (Yonas, 1981), and at 2 months show a decrease in heart rate when placed on the deep side of the drop (revealing interest and processing) but not the shallow side (Bertenthal & Campos, 1990). Infants at 2 months of age also show binocular convergence, and at 5 months reveal sensitivity to kinetic information such as texture accretion and deletion that occurs as objects move closer or farther away (Kellman & Banks, 1998).

Though infants may perceive depth in the first months of life, this does not necessarily mean that this ability and the fear that accompanies refusal to cross the visual cliff are part of a domain-specific "original endowment." For example, the 2-month-olds in the study by Berthental and Campos (1990) perceived the depth of the illusory drop in the visual cliff but showed little or no fear of it. Furthermore, it has been suggested that "experience is the teacher," but not infants' personal experience of falling. Sorce, Emde, Campos, and Klinnert (1985) found that 12-month-olds crossed the deep side of the cliff when their parent smiled and encouraged them to do so; but infants of the same age seldom crossed the same drop when their parent's face showed intense fear. Perhaps, then, infants perceive depth early in life but fear of falling appears considerably later when they become sensitive to and learn about the emotional cues of others or, alternatively, when the relevant adaptive problem is faced.

Shape and Size Perception

Although objects' shape and size remain constant over time, it would be easy for a naive observer to come to a quite different conclusion. As objects move, they often change in orientation or angle, and as they move toward or away from an observer, the retinal image they cast changes. A naive observer could interpret such changes as relating to changes in the shape and size of the objects themselves, and to interpret these changes correctly, an infant (and adult) must perceive shape and size as constant. Seminal work by Bower (1966) suggested that infants as young as 2 months of age possess both shape and size constancy. Bower conditioned infants to turn their head to one stimulus and then presented a number of novel stimuli that varied from that stimulus in orientation, distance, and retinal and real size. Infants responded (with head turns) to objects that were the same shape and size as the original stimulus regardless of their orientation and distance, which suggests that they perceive shape and size as constant.

It is quite possible, however, that 2 months spent observing objects as they move provides plenty of experience for a domain-general learning mechanism to acquire shape and size constancy. Slater and Morrison (1985; see also Slater, Mattock, & Brown, 1990) addressed this issue by testing whether 2-day-old infants also exhibit shape and size constancy in a preferential looking procedure. In one study, newborns' preference for a square (relative to preference for a trapezium presented in a frontoparallel plane) diminished as the orientation of the square moved away from the frontal plane. In a second study, infants were familiarized to changes in slant (thus desensitizing them) for a square or a trapezium. In the test trials, infants were then shown the familiar shape in a novel slant paired with the other, novel shape. All of the infants looked longer at the novel shape, which suggests that the familiar shape was perceived as familiar despite its appearance in a different slant. Similar studies with three-dimensional objects by Slater et al. (1990) provided converging evidence to support these findings, as does research on scanning preferences that showed neonates possess a bias to scan along the horizontal (Haith, 1991). Taken together, then, these studies suggest that even newborns perceive the shape and size of objects as constant.

Why should these aspects of perception be present at birth when depth perception, for example, is not? One possibility is that depth perception is indeed present at birth, but no studies have been sensitive enough to test for it. The finding that infants possess size constancy could be interpreted as meaning they can perceive depth; yet all that is needed for size constancy is to base responses on retinal image size. Another possibility is that shape and size constancy are crucial to survival, even for newborns and very young infants. Why be interested in a human face, or food, if they are perceived to be relatively small? A small morsel of food across the room is unlikely to be as enticing as a large morsel of food that is perceived to be smaller because of its distance from an observer. Alternatively, it is plausible that there is simply insufficient information in the array to specify that objects remain constant in shape and size irrespective of distance and slant. How could a baby learn, for example, that it is an object's retinal image that becomes smaller as that object moves away rather than that the object shrinks in size as it moves away? Regardless, the available data provide compelling evidence that human infants possess an evolved perceptual mechanism that interprets shape and size as constant.

Form Perception

Given that infants are sensitive to shape and size, an important next questions concerns when and how they show sensitivity to the pattern, or form, of objects. Objects are generally composed of a number of features, or parts, and a naive perceiver could interpret these features as unconnected or as bounded together. In fact, it has been suggested that for infants and adults to perceive objects as bound wholes, they must first decompose those objects into a number of parts and then recombine the parts into wholes (Rakison, 2003; Triesman, 1985). This issue has a long history in psychology, going back to

the Gestalt theorists who suggested that infants innately perceive whole forms (Koffka, 1935), as well as theorists with a constructivist viewpoint, who suggested that such perception is learned through experience (Hebb, 1949; Piaget, 1952).

To examine this issue, Slater, Mattock, Brown, Burnham, and Young (1991) conducted an ingenious experiment in which newborns were familiarized simultaneously to stimulus compounds such as a green vertical stripe and a red diagonal stripe. In the test trials, the infants were shown a novel combination of the two elements—for example, a red vertical stripe alongside one of the familiar stimuli. All of the newborns showed a preference to look at the new combination of elements, which suggests that they had processed the color and slant of the line together rather than independently. Again, converging evidence to support this result was found using a similar design that showed that newborns processed two connected lines as a compound—that is, in terms of the angle formed by those lines—rather than independently (Slater et al., 1991; cf. Cohen & Younger, 1984). In conjunction, these studies suggest that infants are born with an evolved predisposition to parse wholes into their component parts, and to recombine them together into a unified whole.

Another important aspect of form perception relates to the ability to categorize objects by forming prototypes. A "prototype" is an average or ideal of a category that can be a specific class (e.g., an average dog is a Labrador) or a more abstract notion (e.g., a square has four equal sides with internal angles of 90 degrees). There is now considerable evidence that infants are able to form perceptual prototypes within the first months of life and do so for a variety of stimuli. In one classic study, Bomba and Siqueland (1983) familiarized 3- and 4-month-olds with distorted shapes (either squares, diamonds, or triangles) and then presented the infants one of the same distorted shapes, as well as the prototypical shape (that was not shown during familiarization). Somewhat counterintuitively, infants looked longer at the distorted shape than at the prototypical shape, which the authors interpreted to mean that they treated the previously unseen prototype as familiar; that is, the prototype did not hold infants' visual attention because it was closer to their representation for the shape than the distorted image. Infants in the first months of life have also been shown to form prototypes for a variety of stimuli, including human faces (Langlois, Roggman, & Musselman, 1994; Rubenstein, Kalakanis, & Langlois, 1999) and schematic animals (Younger, 1985), which could be interpreted as evidence that a prototype formation mechanism is not specific to any particular input and is domain-general. However, as shown in the next section, there is reason to question whether the same perceptual mechanism operates for one class of particularly important stimuli; namely, faces.

Face Perception

If we are experts in nothing else, all adults are experts in the field of face recognition. We can recognize thousands if not tens of thousands of faces; we remember those that were last seen over 30 years ago, and we can identify sex and age from

facial information alone. Does this expertise mean that faces are "special" in some sense and that their perception is driven by an evolved mechanism dedicated to faces? According to one view, this is exactly the correct characterization. Our ancestors needed to recognize specific faces to determine who were enemies, who were friends, and who were family members. They also used facial information in mating to determine health and vitality. And infants who quickly learned the faces of those who protected them and provided for their physical and emotional needs would have been more likely to survive than those who did not. Face expertise, it is therefore argued, is a consequence of the importance of faces in our evolutionary past and is grounded in an innate psychological mechanism that facilitates early orientation to, and learning about, faces (Johnson & Morton, 1991). According to another perspective, however, expertise in face recognition is a product of experience with many faces, and the same domain-general mechanism can be applied, and is applied, to other stimuli. For example, adults with relevant experience are able to make exceptionally fine discriminations between dogs of the same subordinate category (Diamond & Carey, 1986) and are able to sex-type chickens with over 99% accuracy (Gibson, 1969). Clearly, in contrast to the other aspects of perceptual development discussed thus far, the case for or against an evolved face-processing mechanism is far from unequivocal.

Impressive evidence to support the first view was presented in a prominent series of studies by Johnson, Dziurawiec, Ellis, and Morton (1991; see also Goren, Sarty, & Wu, 1975). Newborn infants were presented with stimuli that moved laterally back and forth at about 5 degrees per second, and the experimenters coded the amount of looking and orienting to the stimuli. The stimuli were head-sized paddles that depicted a schematic face (with eyes, eyebrows, a nose, and a mouth), a scrambled face (the same features rearranged in a symmetrical configuration), or a blank face (without any features). Impressively, the newborns looked at and tracked the schematic face more than the scrambled face and the blank face, and they looked at and tracked the scrambled face more than the blank face.

Based on these and related findings (for a review, see Johnson & Morton, 1991), it was proposed that infants are born with two separate but related mechanisms that are specific to information given by faces. The first, labeled CONSPEC, is a subcortical mechanism that causes infants to orient to faces or, at the very least, to face-like structures with eye and mouth elements (Simion, Valenza, & Umilità, 1998). The second, labeled CONLERN, is cortically driven and allows infants to detect and learn about similarities and differences in faces. There are reasons to doubt at least part of this formulation, however. The study by Johnson et al. (1991) provides compelling evidence that newborns track human-like faces, which suggests the existence of something like a CONSPEC mechanism. There is little evidence, though, that infants possess a dedicated face-learning mechanism such as CONLERN over and above a more all-purpose learning mechanism. Furthermore, the claim that any such mechanism is cortically driven is undermined by evidence that

infants discriminate between their mother and a stranger with the first days of life (Walton, Bower, & Bower, 1992). Indeed, Johnson and Morton (1993) acknowledge that CONLERN may be just a domain-general mechanism that "becomes specialized for faces as a result of massive experience of them provided by the species-typical environment" (p. 249). This view is supported by recent studies that examined processing of upright and inverted faces by infants and adults. For example, Pascalis, de Haan, and Nelson (2002; Johnson & de Haan, 2001) found that adults and 9-month-olds fixate visually longer to novel human faces than to familiar human faces but fixate equally long novel and familiar monkey faces. Infants at 6 months of age, however, visually fixate longer to novel than to familiar faces for humans as well as monkeys. The authors suggested that this narrowing of the ability to perceive faces is indicative of cortical specialization, a view that fits well with the idea that infants possess an initial bias for face orientation and, perhaps, recognition that is adapted following experience (Johnson, 2000; Nelson, 2001).

Another line of research provides further support, however, for the claim that newborns possess a dedicated face perception mechanism. In a classic body of work, Meltzoff and Moore (1977, 1994) showed that newborn infants will imitate facial gestures made by an adult, such as tongue protrusion and mouth opening, and by 6 weeks will repeat such gestures 24 hours after seeing them modeled. The basic imitation findings have since been replicated and extended in a number of laboratory conditions, most notably with emotional expressions such as happiness and sadness (Field, Woodson, Greenberg, & Cohen, 1982), and in non-Western cultures such as Nepal (Vinter, 1986). According to Meltzoff and Gopnik (1993), this imitation is possible because of an evolved "like me" mechanism. This mechanism, in combination with cross-modal matching of the visual input (e.g., a tongue protrusion) and the proprioceptive properties of his or her own mouth, leads an infant to draw a comparison between faces he or she observes and the infant's own face. Despite the appeal of such a view, the "like me" hypothesis currently remains controversial, and there is little in the way of direct evidence to support it; indeed, it is unclear how such a theory could empirically be tested. Nonetheless, the findings on imitation by newborns (Meltzoff & Moore, 1977, 1994) provide substantial evidence that infants are born with a predisposition to imitate human facial movements. In all likelihood, such contingent behaviors help to form an emotional bond between caretaker and infant, provide the foundation for communication between them, and may also constitute the first volitional motor activity of the newborn.

A final line of research that is worth noting concerns infants' preference for attractive faces. When infants at 2, 3, and 6 months of age are presented with attractive and unattractive faces (as rated by adults) in a paired-preference procedure, they look longer at the attractive faces than at the unattractive ones and do so for males and females (Langlois et al., 1987). This bias was not, however, a simple preference for symmetry, because infants have been shown to look at vertically symmetrical faces and asymmetrical faces

equally long (Samuels, Butterworth, Roberts, Graupner, & Hole, 1994). Because infants have no experience of what is held by a culture as the standard for attractiveness, these findings could be interpreted as evidence for an evolved preference for attractiveness. According to one view, however, this early preference results from a more general information-processing capacity, namely, prototype formation (Langlois et al., 1987). The reasoning behind this argument is that average, or prototypical (e.g., the distance between facial features is average) faces tend to be highly attractive because humans possess a central tendency preference for prototypical exemplars over nonprototypical exemplars (Halberstadt & Rhodes, 2000). Indeed, adults find a composite face (made by averaging features across multiple faces) more attractive than nearly all of the individual faces from which it was formed (Langlois & Roggman, 1990), and infants as young as 6 months of age show a similar preference, as well as the ability to extract the central tendency from a set of faces (Rubenstein et al., 1999).

To summarize, three lines of research speak to whether infants possess one or more evolved mechanisms designed to facilitate learning of faces. The research on face tracking suggests that newborns orient to faces more than to other stimuli; the research on imitation suggests that newborns repeat facial movements by human faces, and the research on attractiveness suggests that a preference for attractiveness results from prototype formation. Taken together, these findings point to the existence of a bare-boned evolved mechanism for face perception that encourages infants to look at faces and repeat their movements. The findings also suggest, however, that beyond these initial predispositions, face recognition and perception—including the acquisition of information relating to age, gender, and specific exemplars, such as parents—result from more all-purpose, general learning mechanisms.

Movement and Object Perception

The world is primarily a static perceptual field interspersed with movement. Some of this movement has little relevance to the infant, such as the tree moving in the wind or the clouds drifting across the sky. Yet a good deal of movement carries with it important information. Moving entities may be dangerous, potential prey, or they may provide nourishment or protection. And understanding that different object kinds, namely, animates and inanimates, move in different ways is fundamental to reliable prediction about their behaviors. There is considerable evidence that sensitivity to motion is ubiquitous among animal species, and according to one view, this sensitivity has evolved as part of a predator avoidance mechanism: Young vervets, for instance, rely solely on motion cues to categorize potential predators, and will initially make false "eagle" alarm calls for a whole host of birds and even rapidly falling leaves (Evans & Marler, 1995; Seyfarth & Cheney, 1986). An important question is whether young infants are similarly sensitive to motion, how this sensitivity emerges, and how it relates to other aspects of object perception.

A good starting point is the study by Johnson et al. (1991) providing clear evidence that newborns respond to motion and track a moving object. More impressive, perhaps, newborns show a visual preference for moving stimuli over static ones, and they can extract shape information from such stimuli (Slater, 1989). These findings alone suggest that infants possess an evolved predisposition to find movement salient and orient to moving objects. Between 2 and 5 months of age, infants orient to a small moving object in a visual field of static objects (Dannemiller, 2000), and by 3 months of age, infants prefer a human moving point-light display to an unstructured point-light display (Bertenthal, 1993). This sensitivity to movement allows infants to extract information not only about shape but also spatial location. For example, Haith and colleagues (Haith, 1991; Haith, Wentworth, & Canfield, 1993) found that 3-month-olds develop an expectation, or anticipation, about the spatial location of stimuli when shown a repeating left–right pattern.

It might be predicted that it is more difficult for infants to extract information about an object's form when that object is moving, because the array is continuously changing. Similarly, it might be predicted that because objects are often partly occluded by other objects, infants initially perceive the world as made up of many fragments. However, classic research by Kellman and Spelke (1983) on infants' perception of partly occluded objects showed that quite the opposite is true. Infants at 4 months of age were habituated to a static central occluder behind which a rod moved back and forth. Importantly, only the top and bottom of the rod were visible. During the test trials, infants were shown a moving complete rod or two moving separate pieces of rod identical to those seen during habituation. Infants' looking behavior revealed that they interpreted the rod behind the occluder as complete; that is, they visually fixated longer to the incomplete rod than to the complete one. In subsequent studies with a similar design (Slater et al., 1990; Slater, Johnson, Kellman, & Spelke, 1994), newborn infants were found to show the opposite pattern of looking, such that they looked longer at the complete rod than at the incomplete rod. This was the case even if there was a considerable gap between the rod and the occluder (which means that one is definitely behind the other). However, 2-month-olds looked longer at the incomplete rod if the width of the occluder was reduced, so that more of the rod was visible (Johnson & Aslin, 1995; for a review, see Johnson, in press).

According to Kellman (1993; Kellman & Spelke, 1983) these data suggest that common motion, which he refers to as an innate, or maturational "primitive process," is a strong cue that indicates object unity, as does edge alignment. Yet at 4 months of age, neither cue is sufficient, on its own, to specify object unity. Although newborns are sensitive to common motion, it is possible that the fact that two separate pieces of rod (and not one) are seen overrides this dynamic information. However, that young infants require the two rods—or the variety of shapes used in the different experiments on object unity—to be similar suggests that their perception is guided by Gestaltist principles such as *common movement, good continuation* (edges that are unconnected are perceived as connected if the edges are relatable), *proximity*, and

similarity (Kellman & Banks, 1998). These Gestalt principles are presumably an evolved perceptual predisposition that allow even very young infants to perceive the world as consisting of complete objects rather than unconnected fragments. Yet, at the same time, it is evident that experience is required before such principles supersede other available perceptual information.

Causal Perception

To make sense of the world, it is imperative that infants come to understand that actions have causal consequences. This is the case for their own actions, as suggested by Piaget (1952), as well as the actions of others around them. It is now well established that within the first year of life, infants are sensitive to various aspects of causality. Leslie and Keeble (1987), in perhaps the first series of studies to examine systematically this issue, showed 7-month-old infants a series of simple launching events based on those developed by Michotte (1963). Infants were habituated to a green, brick-shaped object that moved from left to right across a screen and contacted a second brick-shaped object (of a different color) that then moved in the same direction until off the screen. In a control condition, they were habituated to similar events, except that there was a short delay between contact and response. In the test phase, infants were shown the same events presented during habituation reversed. In other words, the brick shapes moved from the right side of the screen toward the left side of the screen. The rationale was that the reversal of the launching event changed the relationship between agent and recipient (the agent became the recipient and vice versa), whereas the reversal of the nonlaunching delay event did not. Impressively, 7-month-olds recovered visual attention to the reversal of the launching event more than did infants presented with the reversal of the nonlaunching event.

To explain these and other empirical data, Leslie (1984, 1994, 1995) proposed a nativist and modular view of early causal knowledge whereby infants are born with a three-part theory of agency. (The term "agency" refers to the stable and continuing aspect of objects as agents rather than a single display of agency.) According to Leslie, separate modules in the brain process the mechanical, intentional, and cognitive properties of agents, and it is these modules that direct infants to attend to, and interpret, diverse events in different ways. The perception of spatiotemporal patterns alone does not, according to Leslie, lead infants to interpret an animate as a causal agent; instead, one (or a number) of the causality modules construes spatial and temporal information as an abstract causal code. This is clearly an evolutionary-grounded perspective. The early perception of physical and psychological causality is of fundamental importance to human survival—we must determine which things engage in actions with causal consequences—and domain-specific causal processing mechanisms are an evolved solution to this adaptive problem.

This modular view has been challenged at both a theoretical and an empirical level. Rakison and Poulin-Dubois (2001), for example, pointed out

that it is not clear how different causality modules are "triggered" by the same input, and questioned why infants must possess specific mental structures that draw attention to different kinds of causal information. Take, for instance, a seemingly simple causal event, such as when a human hand reaches for a ball; according to Leslie, such an event involves mechanical and intentional causation. Yet it remains to be seen how information-encapsulated modules work in concert to process such an event. The available empirical evidence has also suggested that causal perception may be learned during ontogeny, and that this learning is grounded in all-purpose information-processing mechanisms. Oakes and Cohen (1990) habituated 6- and 10-month-olds either to direct launching, delayed launching, or noncontact events. The direct launching and delayed launching events were identical to those in the study by Leslie and Keeble (1987). In the noncontact event, infants were habituated to a similar launching sequence except that there was a small gap between the final position of the first object and the starting position of the second object. In the test phase, infants in each condition were presented with the direct, delayed, and noncontact events. The results showed that 6-month-old infants looked equally long at all three test events, which was taken to mean that they did not discriminate causal from noncausal events. In contrast, 10-month-old infants who were habituated to the direct launching event looked longer at the two noncausal events, and those habituated to one of the noncausal events looked longer at the causal launching event but not at the other noncausal event. The authors interpreted these findings to mean that the ability to perceive causality is not automatic within the first year of life but rather emerges between 6 and 7 months of age.

Supporting evidence for this finding was generated in a follow-up study by Cohen and Oakes (1993) using a similar design in which 10- to 12-month-olds failed to discriminate causal launching events from noncausal events when different objects were used in each habituation trial. This finding can be explained within an information-processing perspective. Infants' processing of the different objects in each event meant that they failed to process the causal nature of the events themselves. Such an explanation is highly plausible and suggests that causal perception is not an automatic process that is controlled by evolved specialized modules. Thus, although causal perception is a viable candidate for an evolved psychological mechanism, there is little evidence to support such a possibility. Proponents of an information-processing approach argue that the available evidence points the other way: Causal perception is learned, appearing after 6 months of life, presumably through experience of spatial and temporal contiguities.

Summary of the Evidence for Evolved Mechanisms for Perceptual Development

In this section, I have examined evidence for evolved psychological mechanisms in early perceptual development by evaluating research on depth perception, shape, size, and form perception, face perception, movement and

object perception, and causal perception. Overall, the available evidence indicates that infants may possess a number of evolved mechanisms that facilitate early perceptual development; however, in many cases, they are relatively simple mechanisms that simply help to get the baby's visual perception off the ground. For example, the reviewed data suggests that infants do not need to learn that objects have shape and size constancy, that objects can be closer or farther away, or that relatable edges imply that an occluded object is connected. They are also endowed with a number of *attention biases* (Rakison, 2003) that elevate the salience of particular aspects of the array. Specifically, young infants and newborns attend to motion and to simple face-like stimuli, prefer relatively complex over relatively simple stimuli, and parse simple forms into their component parts and then recombine them into wholes.

There is very little evidence, however, that other aspects of visual perception are underpinned by processes designed by evolution for specific inputs. Infants perceive depth within the first months but do not show fear of a drop until later in life. Infants learn about the specific shapes of objects, circles and squares, for example, as well as about the various faces they encounter through prototype formation and other general learning mechanisms. And infants may learn about physical causal perception—whereby one thing acts on another through contact—via experience with temporal and spatial contiguities. In these, and presumably other aspects of perceptual development not outlined here (e.g., speech perception), there is no need to posit specialized evolved mechanisms to explain how infants come to make sense of the world. In many cases, domain-general processes such as associative learning, prototype formation, conditioning, and imitation, coupled with simple attention biases and other basic perceptual principles—for example, those forwarded by the Gestaltists—can account convincingly for the available data.

COGNITIVE DEVELOPMENT IN INFANCY

The line between cognitive and perceptual development in infancy is often fuzzy. It is frequently hard to delineate where perceptual processes cease and cognitive ones begin. For example, it would be just as feasible to include the coverage of infants' understanding of causality in this section as in the section on perceptual development. As is the case for perceptual development, few researchers have explicitly set out to examine cognitive development from an evolutionary perspective—that is, in terms of adaptations designed to solve a particular problem that faced our ancestors—and, indeed, there are only a handful of studies on adult cognition that have adopted such a stance (cf. Cosmides & Tooby, 1994). Nonetheless, it is indisputable that humans' cognitive abilities distinguish us from other living things. Adult humans are able to remember a huge amount of relatively arbitrary information, solve complicated and multilevel, real-world abstract and mathematical problems, learn and use one or more languages, and form context-dependent categories online

on the basis of surface and deep properties (Barsalou, 1983; Gelman, 2003). Are these abilities evolved specialized processes or modules? Or do they develop out of more general processes and simply reflect the multifaceted and complex world in which we live? To address these questions, at least in part, I examine a moderately small number of domains to assess the role of selection pressures on cognitive development, namely, an understanding of physics, math, and categorization and concept formation for objects.

Infants' Understanding of Simple Physics

To a naive reader, it may seem somewhat outlandish to consider whether infants understand or are sensitive to principles of physics. High school students find it difficult to understand such principles (as do I), so what, one might ask, could infants know about gravity, momentum, and inertia? As far as we know, the laws of physics have remained constant throughout human evolution. According to one view, therefore, it is adaptive for human infants and the young of other species to possess innate knowledge or specialized learning mechanisms that facilitate the ability to learn about such laws early in life. Evidence to support this view is drawn from a number of sources.

Classic work by Spelke, Breinlinger, Macomber, and Jacobson (1992) suggested that young infants appreciate that objects are solid. Two-month-old infants were shown a stage with a ball on one side and a box on the other side, after which a screen was lowered to occlude the box. Infants were then habituated to an event in which a ball rolled behind the screen, after which the screen was raised to show the ball resting against the box. In the test phase, infants were shown two similar events that included a second box placed nearer the ball and separated slightly from the first one. In the "impossible" event, after the screen was raised the ball was shown resting against the farthest box, having seemingly passed through the nearer box. In the "possible" event, the ball was shown resting against the nearest box. Infants visually fixated to the impossible event longer than to the possible event, which the authors interpreted to mean that they understood that one solid object cannot pass through another. A similar conclusion was drawn by Baillargeon and collaborators (Baillargeon, 1987, 1998; Baillargeon, Spelke, & Wasserman, 1985), who found that infants between 3 and 4 months of age looked longer at an event in which a rotating screen appeared to move through the space occupied by a box than at one in which the screen stopped at rest on the box.

Work examining infants' understanding of gravity has shown perhaps even more astounding results. Kim and Spelke (1992), for example, showed that 7-month-olds but not 5-month-olds looked longer at an event in which a ball rolls upward along a slope than at an event in which a ball rolls downward on a slope. The same authors also found that infants at 7 months looked longer at an event in which a ball decreased in speed as it moved down a slope relative to an event in which a ball accelerated in speed as it moved down a slope. Taken together, these findings suggest that toward the middle of the

first year of life, infants start to appreciate that objects are affected by the laws of gravity and momentum. The findings on infants' early understanding of gravity were indirectly extended in a set of studies by Baillargeon and colleagues on the circumstances under which one object was supported by another (Baillargeon, 1998; Baillargeon, Needham, & DeVos, 1992; Needham & Baillargeon, 1993). Infants as young as 3 months of age looked longer at a box that was suspended in midair than at one that fell to the ground. However, it is not until 6 months of age that infants find anomalous a box that is only supported slightly by another box—even if only a small portion of the upper box is in contact with the lower one.

Although both Spelke and Baillargeon have been labeled core-knowledge theorists (Siegler et al., 2003), they have somewhat different views on how knowledge of the physical world is acquired by infants. Spelke suggests that infants are born with a set of innate domain-specific systems of knowledge such as, for example, that two objects cannot be located in the same physical space. Learning, according to this view, is guided by specialized processes that enrich this initial knowledge. Spelke claims that support for this view is the fact that there appears to be some overlap in core knowledge domains in humans and nonhuman animals; that is, "infants' core systems appear to be very similar to those of many nonhuman animals, suggesting that they have a long evolutionary history" (Spelke, 2000, p. 1233). Baillargeon (1995, 1998) similarly argues that infants possess specialized learning mechanisms that facilitate rapid learning in specific domains but does not claim that infants possess innate core knowledge that acts as the foundation for learning.

Once again, however, it is important to examine whether it is necessary to posit that infants are born with evolved domain-specific mechanisms for learning about physics or that they begin life with simple representations about how objects move through space. It certainly seems reasonable to assume that infants in the first 2–3 months of life would observe plenty of unsupported objects fall downward until they contact a horizontal surface. Given the regularity of such events—that objects invariably fall downward—could domain-general processes such as associative learning and habituation account for how infants develop expectations about the way in which objects move according to physical principles? That is, just as Haith et al. (1993) found that 3-month-olds develop an expectation about the spatial location of stimuli that follow a pattern of appearance, so it could be that infants develop an expectation about the motion of unsupported objects following repeated exposure to such events in the real world. A single mechanism that encodes regularities in the world and that operates across multiple modalities and inputs could, at least in principle, account for both sets of findings, as well as numerous others that putatively examine infants' understanding of physics. What of the claim that nonhuman animals possess core knowledge because they share many of the abilities observed in human infants? This too seems like a considerable assumption: It is well known that nonhuman animals learn about their environment in part through habituation, conditioning, associa-

tive learning, and imitation, and it is possible that these mechanisms can account for the similarity in behaviors of human infants and nonhuman animals.

Researchers have recently investigated the findings of Baillargeon with this more domain-general information-processing perspective in mind. These researchers do not directly examine whether domain-specific mechanisms or domain-general mechanisms underpin infants' behavior in tasks on physical principles. Instead, they address whether infants in such tasks are "reasoning" about physical events or responding to simpler aspects of the display. In a series of clever studies by Cashon and Cohen (2000; see also Bogartz, Shinskey, & Schilling, 2000) 8-month-olds' perception of object solidity and object permanence was tested in an extension of the rotating-screen studies discussed earlier (e.g., Baillargeon, 1987). Infants were habituated to one of four events on a computer screen and then tested with all four events. Some of the events were identical to those used by Baillargeon (e.g., a screen that rotated through either 180 degrees or 120 degrees without a block), and others were different (e.g., a screen that rotated through either 180 degrees or 120 degrees with a block sitting in its path). Infants' looking times suggested that they responded on the basis of perceptual novelty—that is, their visual fixation was affected by what changed between habitation and test—rather than on the basis of the possibility or impossibility of the events. Similarly, Rivera, Wakely, and Langer (1999) found that infants have a preference for the 180 degree rotation of the screen over the 120 degree rotation, which suggests that they were not reasoning about physically impossible object permanence events in the studies by Baillargeon and colleagues. The interpretation of the studies by Spelke et al. (1992) have also been questioned by Cohen, Gilbert, and Brown (1996), who found that, until 10 months of age, infants showed no understanding that one object cannot pass through another.

These studies suggest that it is not necessary to impute infants with an evolved specialized mechanism that facilitates early and rapid learning about physical principles. They bring into question whether infants are "little scientists" who reason about physical events by applying innate knowledge, or who possess innate mechanisms that allow them to learn about such events early in life. In contrast, they suggest that infants' behavior can be explained by their response to the introduction or deletion of aspects of the visual display. It is important to note, however, that, without exception, these studies rely on computer-presented animated events rather than real ones, and proponents of the core-knowledge view have suggested that infants process real events and animated computer-presented events differently. At the same time, the theoretical challenge that infants develop expectations about regularities in the world via domain-general processes suggests the possibility that they do not learn about physical events through specialized cognitive mechanisms. Unfortunately, it is unclear whether this issue can be resolved, because newborns and very young infants do not have the requisite perceptual abilities to view relatively complex events such as those in the studies by Spelke and

Baillargeon. Research continues at a brisk pace in this area, and the ever-growing database on infants' physical understanding promises to provide at least some answers to these issues.

Infants' Understanding of Mathematics

The study of infants' interpretation of physical events led researchers to consider other aspects of early science-like reasoning. Most notable among these areas was research on simple aspects of mathematics, such as addition and subtraction. Since the pioneering work of Piaget (1952) on conservation, developmentalists have generally assumed that young children, never mind infants, lack simple, as well as abstract, numerical abilities because, among other things, they do not possess the requisite representational structures. There is, however, considerable evidence that animals are capable of counting as high as 24 and perhaps considerably higher. For instance, rats will press a lever a specific number of times for numbers between 4 and 24 to obtain food (Platt & Johnson, 1971), and parrots and chimpanzees are able to combine numerical symbols in mathematically appropriate ways (Boysen & Berntson, 1989; Pepperberg, 1987). Indeed, perhaps spurred on in part by such work with nonhuman animals, researchers have come to believe that "more and more it appears that infants are capable of reasoning about the effects of the arithmetic operations of addition and subtraction" (Gelman & Williams, 1998, p. 588).

In one already classic study, Wynn (1992) habituated 5-month-olds to a single toy animal on a stage, which was then occluded by a screen. A human hand then placed a second, identical toy behind the screen. The screen was then removed to reveal either one or two toys (on alternating trials). The rationale for this design was that if infants did not represent the two objects over occlusion, they would be expected to fixate visually at the display of two toys longer than at the display of one toy, because only one toy was visible at a time. In contrast, if infants represented each object separately behind the occluder, that is, 1 + 1 = 2, they would be expected to fixate visually longer at the display containing one toy than at the one containing two toys. In fact, Wynn found that infants showed the latter looking preference. Using a similar procedure, Wynn (1992) habituated infants to events in which two objects were shown and then occluded, after which one was removed from behind the screen. Infants in this study looked longer at the test event with two toys than at the event with one toy. This was interpreted to mean that they had represented the two toys behind the occluder and successfully computed the subtraction of two minus one.

In ensuing studies, moreover, infants who were presented with the task of adding one object to another looked longer at three objects than at two objects (Wynn, 1992). These basic findings have since been replicated and extended in a number of ways. Koechlin, Dehaene, and Mehler (1998) showed that infants responded appropriately to the number of objects even

when the occluded objects moved on a turntable, which suggests that they responded to object number and not simply location. More impressive, perhaps, Simon, Hespos, and Rochat (1995) found that infants responded in a similar fashion even when the identity of the object changed while it was behind the screen, which could be taken to mean that they represented the number of occluded objects and not properties the objects had in common, such as their color or shape.

According to Wynn (1992, 1995), these findings suggest that infants possess an evolved innate system of numerical knowledge. Although such results are impressive, it is important to ask what adaptive problems simple numerical abilities such as addition and subtraction help infants to overcome. One can imagine that adults in our evolutionary past must have had to keep track of the number predators or prey, food, and even children in their environment. However, it is not so obvious why it is important or necessary for infants to keep track of such things. One possible answer concerns food; infants who discriminate one morsel of food from two morsels of food might opt to obtain the latter rather than the former and thus obtain more nourishment. Indeed, recent work by Feigenson, Carey, and Hauser (2000) showed that 8- to 12-month-old infants are more likely to crawl toward three cookies than toward two cookies. There is a difference, though, between discrimination of one from two—which can be guided by simple perceptual processes—and a numerical competence that supports addition and subtraction. Furthermore, infants' understanding of the effects of addition and subtraction are limited to small numbers, such that their competence is displayed only when the total number of objects is three or fewer.

These lines of reasoning have ignited a considerable body of research and theory that brings into doubt the findings by Wynn (1992). Most notably, Haith and Benson (1998; see also Clearfield & Mix, 1999) suggested that infants' behavior in the studies by Wynn and others is guided by a perceptual process called *subitizing* rather than numerical reasoning. Subitizing, they suggest, allows infants, children, and adults who observe a small number of items—namely, three or fewer—to immediately encode how many are present. A representation of the number of items is formed, along with any other items that are added or subtracted, and infants will therefore look longer at an image that contains more or fewer items than is stored in their representation. A corollary of this view is that infants will find it difficult to "add" or "subtract" in circumstances when it is more difficult to form a mental representation of the items, and research with 5-month-olds suggests that this is indeed the case (Uller, Carey, Huntley-Fenner, & Klatt, 1999). Although this argument is premised on the idea that infants are not calculating when they add or subtract, because the processes involved are inherently perceptual, it is important to bear in mind that subitizing is very much a specialized, and presumably evolved, mechanism for dealing with numbers.

Other researchers have evaluated the findings of Wynn (1992) and others from more of an empirical perspective. Cohen and Marks (2002) hypothe-

sized that instead of reasoning about addition and subtraction, infants looked longer at events in the Wynn (1992) procedure because of a preference for familiarity in conjunction with a tendency to look longer when more items were presented. To test this idea, Cohen and Marks (2002) first replicated the basic Wynn (1992) study. In subsequent experiments, one group of 5-month-old infants was presented with the test events without exposure to the addition or subtraction habituation events, and another group was familiarized to either one or two items before each test trial, again without showing any addition or subtraction events.

The pattern of looking across the experiments supported the authors' explanation of the infants' behavior in studies on numerical competence; that is, infants tended to look longer at familiar arrays—having seen one object prior to the test phase, infants preferred to look at one object during the test phase—and they showed a general preference to look longer at more items than at fewer items (e.g., two rather than one). As one might expect given the contentious nature of claims about numerical competence in infants, the design and interpretation of the Cohen and Marks (2002) experiments have drawn criticism from a number of researchers on both sides of the fence (Carey, 2002; Mix, 2002; Wynn, 2002). Nevertheless, such studies do suggest that, as has been stressed throughout this chapter, it is necessary to be wary of a rich interpretation of infants' behavior in tasks that claim to show precocious cognitive abilities. In light of the issues under consideration, and in combination with the subitizing theory forwarded by Haith and Benson (1998), they also suggest that it is premature to conclude that infants possess an evolved specialized system that is dedicated to numerical reasoning. Indeed, as I suggested earlier in this section, it is unclear how and why such a system would have evolved in infants. That work in this area began relatively recently suggests that the answers to these concerns may yet be forthcoming; however, as things stand, it seems prudent to conclude that there is little evidence to support the view that humans possess an innate numerical competence system.

Categorization of Objects and Entities in Infancy

The ability to categorize, or group together things as the same, is perhaps the most fundamental of all the cognitive processes. It acts as the foundation for, among other things, language, problem solving, induction, and memory, and without it, the world would seem a chaotic and overwhelming place. Categorization, more specifically, is the process whereby discriminable properties, objects, or events are grouped into classes by means of some principle or rule. To categorize would be to treat, for example, various animals as equivalent and at the same time as different from various plants. The mental representation that encapsulates the qualities and features among items within categories, as well as the structure that exists across categories (e.g., dogs belong to the category of animals) is commonly referred to as a *concept*. Categories,

then, tend to be sets of things in the world, whereas concepts are the internal mental representation of those sets (Margolis, 1994; Smith, 1995).

Clearly, it is crucial for young infants' survival to categorize the world appropriately. They must determine, for example, which objects can be acted upon and which things cause other things to act, which things are dangerous, and which things offer nourishment. Early categories must also constitute the foundation for those used in later life. Among other things, adults categorize people on the basis of their sex and abilities, animals on the basis of whether they are predators, prey, or companions, and objects on the basis of their function. Veridical categories and concepts of the world are therefore necessary not only to survive but also to reproduce. There is little debate that categorization in early infancy is fundamentally perceptual; that is, infants group together a variety of things on the basis of their surface appearance by attending to individual features, correlations among features, and by encoding the form of individual exemplars from which are produced prototypes (Bomba & Siqueland, 1983; Rubenstein et al., 1999). At the same time, however, theorists differ considerably in their view of the mechanisms for learning that allow infants to go beyond information given in the perceptual input. It is not enough for infants, and older children, to learn about only the surface features or shapes of objects: At some point in time, it is necessary to acquire knowledge about the motion properties of objects and entities (e.g., whether they act as agents or are self-propelled), their psychological properties (e.g., whether they are goal-directed or are potentially dangerous), and ultimately their internal biological or mechanical properties (e.g., whether they have a heart or an engine).

According to one group of theories, the acquisition of knowledge about such *nonobvious* properties is underpinned by evolved innate processes or structures (e.g., Leslie, 1995; Mandler, 1992; Gelman, 1990; Premack, 1990). The details of these theories vary somewhat, yet they share the view that infants, particularly those in the first year of life, quickly acquire knowledge about the nonobvious properties of objects and entities, and rely on this knowledge in their category membership decision. Leslie's (1984, 1995) theory of early concept development, which was outlined earlier in the chapter, is perhaps one of the most modular of these views. A similar, though less modular, perspective was proposed by Rochel Gelman (1990; Gelman, Durgin, & Kaufman, 1995). Gelman posited that infants are rapidly able to form concepts of animate entities and inanimate objects that include information about their nonobvious properties because of innate *skeletal causal principles*. These skeletal principles guide infants' attention to aspects of the type of motion and composition—that is, their energy source and materials—of objects and entities, and allow such properties to act as the basis for categorization. A related and equally nativist view was offered by Premack (1990), who suggested that specialized modules cause the perception of self-propelled objects that display goal-directed behaviors to be interpreted as intentional and to be assigned to the psychological domain.

Mandler (1992, 2003) presented one of the most prominent theories of category and concept development that is centered on the notion of an evolved and innate specialized process. According to Mandler, the earliest categories are perceptually based and are represented in terms of a prototype. She also suggests, however, that by 12 months of age or even earlier, an innate specialized process called *perceptual analysis* allows infants to form image schemas that summarize crucial characteristics of objects' spatial structure and movement. These image schemas are posited to recode the perceptual display into a more abstract but at the same more accessible format (Mandler, 1992). Once such information is abstracted, infants will no longer categorize objects on the basis of their appearance but instead group objects together because of a shared *meaning*. Thus, at 3 months of age, infants categorize cats as the same due to a perceptual prototype that specifies which feature values belong to that class (Quinn, et al., 1993), but at 12 months, infants categorize cats as the same because they are self-propelled entities that move nonlinearly.

Evidence to support this theory is drawn from three distinct, infant-related experimental procedures. In particular, Mandler (Mandler & McDonough, 1993, 1996, 1998; Mandler, Bauer, & McDonough, 1991) has used the object manipulation paradigm, in which infants touch sequentially the objects they consider to be the same; the object-examining paradigm, which is an object-based habitation task; and the inductive generalization paradigm, in which infants observe an action and imitate it with one of two test objects. Results from the first two procedures revealed that infants from 12 months of age form categories even when stimuli possess little within-class perceptual similarity—for example, animals, vehicles, or furniture (Mandler et al., 1991)—and by 9 months of age, they categorize as different perceptually similar items such as planes and birds (Mandler & McDonough, 1993). The results of the induction procedure are perhaps more surprising: Infants as young as 9–11 months extend behaviors such as drinking and sleeping to animals and behaviors such as "starting with a key" and "giving a ride" to vehicles.

The views just described have a common perspective: Infant category and concept development requires some kind of innate representational structure or mechanism that has evolved over phylogeny. Once again, however, before accepting this claim it is necessary to consider the nature of the input and whether the available data provide acceptable evidence. Regarding the nature of the input, a number of researchers have suggested that categories in the world are tightly structured not only in terms of their surface appearance but also their deeper, nonobvious properties. For example, dogs tend to look alike, both in terms of overall shape and the specific features they possess, move in similar ways, exhibit similar behaviors (e.g., wag their tails when happy, bark), and have similar internal properties (e.g., brains, dog DNA). Thus, it is argued, domain-general processes—such as associative learning—that are sensitive to regularities in a richly structured input will be sufficient to acquire concepts about surface properties, as well as deeper, nonobvious ones

(Jones & Smith, 1993; Rakison, 2003; Rakison & Hahn, in press; Smith, Colunga, & Yoshida, 2003). In support of this idea, there is substantial evidence that infants in the first and second year of life are sensitive to statistical regularities in the visual, as well as auditory, domains (e.g., Rakison & Poulin-Dubois, 2002; Saffran, Aslin, & Newport, 1996). Indeed, even preschoolers attend to such information in category membership decisions involving surface and nonobvious features (e.g., lives in trees, has a round heart) (Rakison & Hahn, in press).

On a related theme, a number of researchers have questioned the need to invoke abstract conceptual knowledge about objects' meaning in order to explain the available data (Quinn & Eimas, 2000; Quinn, Johnson, Mareschal, Rakison, & Younger, 2000; Rakison, 2003). I focus here on the evidence put forward by Mandler and colleagues (Mandler et al., 1991; Mandler & McDonough, 1993, 1996, 1998), because it constitutes the most substantial and influential body of work to date that is used to support the notion of a specialized mechanism for category and concept formation. One challenge to these data revealed that infants' category formation is grounded in surface features rather than in an understanding of the meaning of objects. Rakison and Butterworth (1998; Rakison & Cohen, 1999) found that object parts, such as legs and wheels, act as the basis for 14- to 18-month-old infants' superordinate category formation (e.g., animals, vehicles, and furniture). For example, infants categorize animals and furniture with legs as equivalent but as different from vehicles with wheels, but they do not categorize animals as different from vehicles when animals possess no legs and vehicles possess no wheels. Similarly, research with the inductive generalization procedure showed that infants' early knowledge of object motion is closely linked to object features. For example, if an experimenter models a cat walking, an infant will imitate this action with a dog and a table—which both have legs—but not with another animal without legs (e.g., a dolphin) (Rakison, in press).

These findings cast doubt over the presence of evolved psychological mechanisms for category and concept development. They suggest that all-purpose learning mechanisms and a sensitive perceptual system are adequate to account for a wide range of empirical data. Domain-general processes are sufficiently robust and sensitive for infants to acquire knowledge from the complex but richly structured input. There are some domains for categorization, however, in which the input is not so richly structured, and in which the cost of relatively slow concept formation may be high. One such domain that is probably more crucial to early survival than those described earlier (e.g., concept formation for animates and inanimates) is that of predator avoidance. The evolutionary psychology perspective suggests that it is more adaptive for predator detection processes, at least, to be innate, because it is not efficacious to learn which animals are dangerous through an encounter in a potentially life-threatening situation. Evidence to support this view is widespread in the animal behavior literature. For example, it has been suggested that the rabbit

retina has a "hawk" detector (Marr, 1982), and salamanders bred from larvae show a predator response—by moving to shelter—in the presence of chemical cues of predatory fish (Kats, Petranka, & Sih, 1988). Perhaps more impressive, toad tadpoles show an antipredator response to three species that prey upon them—backswimmers, waterbugs, and garter snakes—but not to two potential predators that find them unpalatable (trout and newts) (Kiesecker, Chivers, & Blaustein, 1996).

With respect to the relevance of predator detection and avoidance mechanisms in humans, it is notable that the most common object-related phobias in adults are of snakes and spiders. Although the traditional view of phobic anxiety has generally relied on the concepts of fear conditioning (e.g., the pairing of a neutral stimulus with a negative event), more recent conceptualizations of phobias have taken evolutionary psychology into consideration. Work with adults, for instance, revealed that conditioned fear is more resistant to extinction with fear-relevant than with fear-irrelevant stimuli (Öhman & Mineka, 2001). Moreover, in one study by Mineka, Davidson, Cook, and Keir (1984), humans and nonhuman primates were conditioned to conspecifics exhibiting fear to dangerous items with evolutionary relevance (e.g., snakes) and nondangerous items (e.g., flowers). Impressively, both the humans and nonhuman primates acquired fear more rapidly and readily for the dangerous items than for the nondangerous items. This is not to say, however, that fears emerge in the absence of experience. Presumably, humans and nonhumans need to experience the fear exhibited by conspecifics in the presence of particular stimuli; however, there appears to be a bias to learn fears for some stimuli over others.

Based on this evidence, there is reason to suppose that infants, too, may be born with some kind of minimal, information-specific predisposition for things that were dangerous throughout human evolutionary history (for a review, see Barrett, in press). One indirect example of such a disposition is the classic work by Ainsworth et al. (1978) showing that when an unfamiliar adult, and in particular a male adult, approaches an infant, that infant is likely to show fear. Notably, this response is rarely shown toward small children, which indicates that infants' anxiety around strangers may result from a form of "predator" avoidance, whereby adults who are not familiar are treated as potentially dangerous. To examine this issue more directly, I recently gathered preliminary data that suggest that the basic perceptual structure of snakes and spiders may be specified early in life (Rakison, 2001). Infants at 10 months of age were familiarized with five different snakes or five different spiders. During the test phase, infants were shown a familiar predator (e.g., a snake, if familiarized to snakes), a novel predator (e.g., a spider, if familiarized to snakes), and a nonpredator (a frog or rabbit). Infants' looking times revealed that they recovered attention to the nondangerous animal but not to the novel or familiar dangerous animals; that is, infants looked equally long at the novel snake and novel spider, and they looked significantly longer at the frog or rabbit. It is important to stress that these results are tentative, and studies with younger infants and with moving images are currently under way. Note also

that infants showed no external fear response—measured by affective behavior—to the stimuli, and that parents stated that their infants had little or no exposure to the various stimuli. The implication of this finding is that infants may have an innate representational construct that allows them to recognize predators, but the appropriate external fear response does not emerge until later.

In summary, although a number of theorists have suggested that innate specialized mechanisms are required for infants to learn about the complex categories of the world, the available evidence does not provide support for this view. Indeed, there is considerable reason to believe that domain-general processes are sufficient for infants to learn about the objects and entities around them. At the same time, there is good reason to believe that human infants, as with the young of other species, may have some specific predisposition for certain animals that have posed a threat to survival in the course of evolution. Initial evidence suggests that snakes, spiders, and adult humans may fall under this heading, though further research is needed to establish firmly whether indeed this is the case.

Summary of the Evidence for Evolved Mechanisms for Cognitive Development

In this section, I have examined evidence for evolved psychological mechanisms in early cognitive development for a naive understanding of physics, numerical reasoning, and object concept and category formation for animate entities and inanimate objects. In contrast to the review of the perceptual development literature, there is little incontrovertible evidence and considerably more debate over the existence of specialized cognitive mechanism in early development. Coverage of the literature on all three areas of early cognition revealed that there are both theoretical and empirical challenges to the idea that infants possess such mechanisms. At a theoretical level, I propose that domain-general processes are sufficient to encode the rich and consistent structure of the input in the domains of physics, mathematics, and categorization. It is unclear why specialized psychological mechanisms would have evolved for domains in which there are structured regularities that have existed over evolutionary time and that could be readily encoded by more general, all-purpose learning mechanisms. For example, all-purpose mechanisms such as habituation, conditioning, and associative learning can account for learning that unsupported objects fall until contacting a horizontal surface, that 1 + 1 is always 2, and that things with wings tend to fly; that is, given the (albeit brief) review of the literature presented here, it is far from clear that infants possesses (or need to possess) domain-specific mechanisms that prepare them more rapidly to learn about information in these areas.

At an empirical level, a number of researchers have presented data that directly contradict the interpretation that infants are precocious cognizers. For example, studies by Baillargeon and colleagues (Baillargeon, 1987, 1998;

Baillargeon et al., 1985) revealed putatively that infants understand that one object cannot pass through another object, yet several researchers have shown that perceptual preferences can account for such findings (Bogartz et al., 2000; Cashon & Cohen, 2000; Rivera et al., 1999). Similarly, Wynn's (1992) work suggested that in the first year of life, infants are capable of addition and subtraction for small numbers; yet Cohen and Mark's (2002; see also Haith & Benson, 1998) research implied that infants' behavior in numerical tasks is driven by perceptual and attention biases. Finally, Mandler's (Mandler et al., 1991; Mandler & McDonough, 1993, 1996, 1998) work suggested that infants rely on abstracted knowledge in categorization and induction decisions, but research by Rakison and collaborators (Rakison, in press; Rakison & Butterworth, 1998; Rakison & Cohen, 1999) indicated that the initial basis for such behavior is surface features.

There is, therefore, little unequivocal evidence that infants possess evolved cognitive mechanisms for learning about physics and simple mathematics, or for category and concept formation. Perhaps, this is not surprising given the later development of cognitive abilities relative to perceptual ones; that is, it is difficult to gather such data for abilities that emerge considerably after birth. At the same time, however, I have suggested that one domain in which one is more likely to find evidence for such mechanisms is that in which there is a high cost for failure to acquire the relevant information quickly and the input is not too richly structured. One such domain relates to predator detection and recognition, but conceivably research on other important areas relating to food and protection may impart similar findings.

CONCLUSIONS

In this chapter, I have reviewed a broad literature on early perceptual and cognitive development. The evaluated literature focused on perceptual and cognitive development in the visual modality but at the same time spanned a broad range of areas, including depth perception, shape, size, and form perception, face perception, movement and object perception, causal perception, naive physics and mathematics, and object category and concept formation. The overarching aim of the review was to determine whether there is evidence in these domains of inquiry for evolved, specialized psychological mechanisms or more general, all-purpose ones. Those who adopt an evolutionary perspective (e.g., Buss, 2004; Cosmides & Tooby, 1994) typically highlight the former as the adaptive solution to the problems faced by our hominid ancestors, while at the same time rejecting the idea that there exists any "general adaptive problem" (Symons, 1992). Thus, evidence that specialized, presumably modular mechanisms are present early in life and facilitate perceptual or cognitive development in particular domains could be interpreted as support for this view.

The review of the infant perceptual development literature revealed that infants may indeed possess evolved mechanisms that allow them to perceive shape and size as constant, depth, and occluded objects as connected. It was also suggested that infants are born with perceptual *attention biases* that cause them to attend to specific facets of the array. Included in these attention biases are a preference for dynamic over static cues, simple face-like stimuli, and complex rather than simple stimuli. There is also evidence that, across a range of stimuli, newborns and young infants process whole forms by breaking them down into their component features or dimensions. The review of the infant cognitive development literature was not as forthcoming in providing tangible evidence of evolved, specialized mechanisms that allow infants to learn about the world around them. Although a number of researchers claim that data on early physical understanding, mathematical reasoning, and category and concept development support, directly or indirectly, the evolutionary perspective (e.g., Baillargeon, 1987; Mandler & McDonough, 1993, 1998; Spelke et al., 1992; Wynn, 1992), in many cases, there is counterevidence suggesting that the data can be explained by more all-purpose mechanisms or perceptual preferences. Indeed, the review demonstrated that although a small number of evolved processes play an important role in the earliest aspects of perceptual and, perhaps, cognitive development, the brunt of the work in these domains is performed by more general mechanisms that operate across a range of domains.

As outlined earlier, I suggest that there is a clear pattern in these findings. Infants are endowed with a number of basic perceptual abilities, such as size constancy, as well as some perceptual preferences. These abilities and preferences, in conjunction with all-purpose mechanisms for learning, are sufficient for those domains in which there is a rich structure in the input and little survival cost in a relatively slow developmental time table. Attention to motion, coupled with size and shape constancy, and associative learning can explain, for example, how infants learn that unsupported objects will fall until they contact a horizontal surface. Similarly, associative learning, coupled with a perceptual bias to attend to specific object features and the motion characteristics of objects, can account for how infants learn about the way that animates and inanimates move in the world.

At the same time, the same underlying mechanisms are not adaptive for domains in which there is comparatively little structure in the input and/or those domains in which there is a high survival cost in a slow developmental timetable. For example, infants' ability to survive is contingent on recognizing the faces of caretakers, forming emotional bonds with them, and avoiding strangers. An evolved face-orienting mechanism and an innate wariness of unknown adults provides an important foundation for these behaviors. Likewise, infants must avoid dangerous predators and steep drops, and they cannot learn about the threat these things pose through experience. Infants may therefore have a minimal information-specific predisposition for snakes and

spiders, as well as drops over a certain height. In both cases, however, experience is necessary to trigger the emotion or response that is appropriate for these contexts. Information-specific predispositions do not automatically come online at some point in development; rather, a small number of experiences with those events, and perhaps of conspecifics' reaction to them, is required.

Despite the fact that the review supports the idea that domain-general processes underlie a considerable amount of perceptual and cognitive development, it is important to note that these all-purpose mechanisms are adaptive solutions to the problems that infants have faced throughout human evolutionary history. This is not a popular view among evolutionary psychologists and will most likely be berated by those who adopt a more hardline evolutionary perspective. Yet it is impossible to dismiss the existence of such all-purpose mechanisms in humans and nonhuman animals, and the issue has been, and will remain, to what extent these mechanisms are responsible for our representation of the perceptual and cognitive world. I suggest that it is entirely feasible for domain-specific and domain-general mechanisms to have evolved alongside each other (for similar proposals, see Bjorklund & Pellegrini, 2002; Geary & Huffman, 2002). In light of the ubiquity of domain-general mechanisms in the animal kingdom, it is likely that they were the earliest learning apparatus to evolve. Subsequent species-general adaptive problems would cause the emergence of species-general mechanisms such as depth perception, and species-specific adaptive problems would cause the emergence of species-specific mechanisms such as face orienting in humans.

Finally, it is worth noting that it is assumed that the mechanisms involved in early perceptual and cognitive development continue to play a fundamental role in perception and cognition throughout the lifespan. However, although some of the early cognitive-developmental mechanisms remain (e.g., associative learning), many are usurped by a more "advanced" level of cognitive processing that emerges through more explicit learning (e.g., mathematical reasoning). Thus, although infants across many cultures may start with the same evolved, domain-specific and domain-general mechanisms, it is entirely probable that as infants reach preschool age, there will be an observed developmental divergence in their cognitive and, less so, their perceptual abilities. The role of language and formal tuition will account for a good deal of this divergence (see e.g., Smith et al., 2003), but are unlikely to affect the basic learning mechanism that evolution has endowed upon the child.

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